# Question ID: f224df07

**Question**

A cargo helicopter delivers only 100-pound packages and 120-pound packages. For each delivery trip, the helicopter must carry at least 10 packages, and the total weight of the packages can be at most 1,100 pounds. What is the maximum number of 120-pound packages that the helicopter can carry per trip?

**Answer**

A. 2

B. 4

C. 5

D. 6

# Question ID: 64c85440

**Question**

In North America, the standard width of a parking space is at least 7.5 feet and no more than 9.0 feet. A restaurant owner recently resurfaced the restaurant's parking lot and wants to determine the number of parking spaces, n, in the parking lot that could be placed perpendicular to a curb that is 135 feet long, based on the standard width of a parking space. Which of the following describes all the possible values of n ?

**Answer**

A. $18 \leq n \leq 135$

B. $7.5 \leq n \leq 9$

C. $15 \leq n \leq 135$

D. $15 \leq n \leq 18$

# Question ID: 968e9e51

**Question**

$y \leq x$

$y \leq -x$

Which of the following ordered pairs $(x,y)$ is a solution to the system of inequalities above?

**Answer**

A. $(1,0)$

B. $(-1,0)$

C. $(0,1)$

D. $(0,-1)$

# Question ID: bf5f80c6

**Question**

$$y < -4x + 4$$

Which point $(x, y)$ is a solution to the given inequality in the $xy$-plane?

**Answer**

A. $(-4, 0)$

B. $(0, 5)$

C. $(2, 1)$

D. $(2, -1)$

# Question ID: 80da233d

**Question**

A certain elephant weighs 200 pounds at birth and gains more than 2 but less than 3 pounds per day during its first year. Which of the following inequalities represents all possible weights w, in pounds, for the elephant 365 days after birth?

**Answer**

A. $400 < w < 600$

B. $565 < w < 930$

C. $730 < w < 1{,}095$

D. $930 < w < 1{,}295$

# Question ID: b31c3117

**Question**

$H = 120p + 60$

The Karvonen formula above shows the relationship between Alice's target heart rate H, in beats per minute (bpm), and the intensity level p of different activities. When $p = 0$, Alice has a resting heart rate. When $p = 1$, Alice has her maximum heart rate. It is recommended that p be between 0.5 and 0.85 for Alice when she trains. Which of the following inequalities describes Alice's target training heart rate?

**Answer**

A. $120 \leq H \leq 162$

B. $102 \leq H \leq 120$

C. $60 \leq H \leq 162$

D. $60 \leq H \leq 102$

# Question ID: 6cb9bf45

**Question**

$$y > 7x - 4$$

For which of the following tables are all the values of $x$ and their corresponding values of $y$ solutions to the given inequality?

**Answer**

A.

| $x$ | $y$ |
|---|---|
| $3$ | $13$ |
| $5$ | $27$ |
| $8$ | $48$ |

B.

| $x$ | $y$ |
|---|---|
| $3$ | $17$ |
| $5$ | $31$ |
| $8$ | $52$ |

C.

| $x$ | $y$ |
|---|---|
| $3$ | $21$ |
| $5$ | $27$ |
| $8$ | $52$ |

D.

| $x$ | $y$ |
|---|---|
| $3$ | $21$ |
| $5$ | $35$ |
| $8$ | $56$ |

# Question ID: 0ef4a7b6

**Question**

A particular botanist classifies a species of plant as tall if its typical height when fully grown is more than $100$ centimeters. Each of the following inequalities represents the possible heights $h$, in centimeters, for a specific plant species when fully grown. Which inequality represents the possible heights $h$, in centimeters, for a tall plant species?

**Answer**

A. $106 < h < 158$

B. $80 < h < 100$

C. $42 < h < 87$

D. $17 < h < 85$

# Question ID: 6daf8d70

**Question**

A city employee will plant two types of bushes, azaleas and boxwoods, in a park. There will be no more than $164$ total bushes planted, and the number of azaleas planted will be at most three times the number of boxwoods planted. Which of the following systems of inequalities best represents this situation, where $a$ is the number of azaleas that will be planted, and $b$ is the number of boxwoods that will be planted?

**Answer**

A. $a + b \geq 164$

$3a \geq b$

B. $a + b \geq 164$

$a \leq 3b$

C. $a + b \leq 164$

$3a \geq b$

D. $a + b \leq 164$

$a \leq 3b$

# Question ID: 948087f2

**Question**

$$y \leq 3x + 1$$

$$x - y > 1$$

Which of the following ordered pairs (x, y) satisfies the system of inequalities above?

**Answer**

A. $(-2, -1)$

B. $(-1, 3)$

C. $(1, 5)$

D. $(2, -1)$

# Question ID: 90bd9ef8

**Question**

The average annual energy cost for a certain home is $4,334. The homeowner plans to spend $25,000 to install a geothermal heating system. The homeowner estimates that the average annual energy cost will then be $2,712. Which of the following inequalities can be solved to find t, the number of years after installation at which the total amount of energy cost savings will exceed the installation cost?

**Answer**

A. $25{,}000 > (4{,}334 - 2{,}712)t$

B. $25{,}000 < (4{,}334 - 2{,}712)t$

C. $25{,}000 - 4{,}334 > 2{,}712t$

D. $25{,}000 > \frac{4{,}332}{2{,}712}t$

# Question ID: e9ef0e6b

**Question**

A model estimates that whales from the genus *Eschrichtius* travel $72$ to $77$ miles in the ocean each day during their migration. Based on this model, which inequality represents the estimated total number of miles, $x$, a whale from the genus *Eschrichtius* could travel in $16$ days of its migration?

**Answer**

A. $72 + 16 \leq x \leq 77 + 16$

B. $(72)(16) \leq x \leq (77)(16)$

C. $72 \leq 16 + x \leq 77$

D. $72 \leq 16x \leq 77$

# Question ID: 74c98c82

**Question**

An event planner is planning a party. It costs the event planner a onetime fee of $\$35$ to rent the venue and $\$10.25$ per attendee. The event planner has a budget of $\$200$. What is the greatest number of attendees possible without exceeding the budget?